MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, PRINTED WIRING BOARD RECEPTACLE, CARD INSERTION, CONTACT SPACING (.125) TYPE AD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-21097.

TYPE AD CONNECTORS SHALL NOT BE USED IN AIRBORNE EQUIPMENT.

NOT FOR USE IN NEW DESIGNS OF ARMY EQUIPMENT

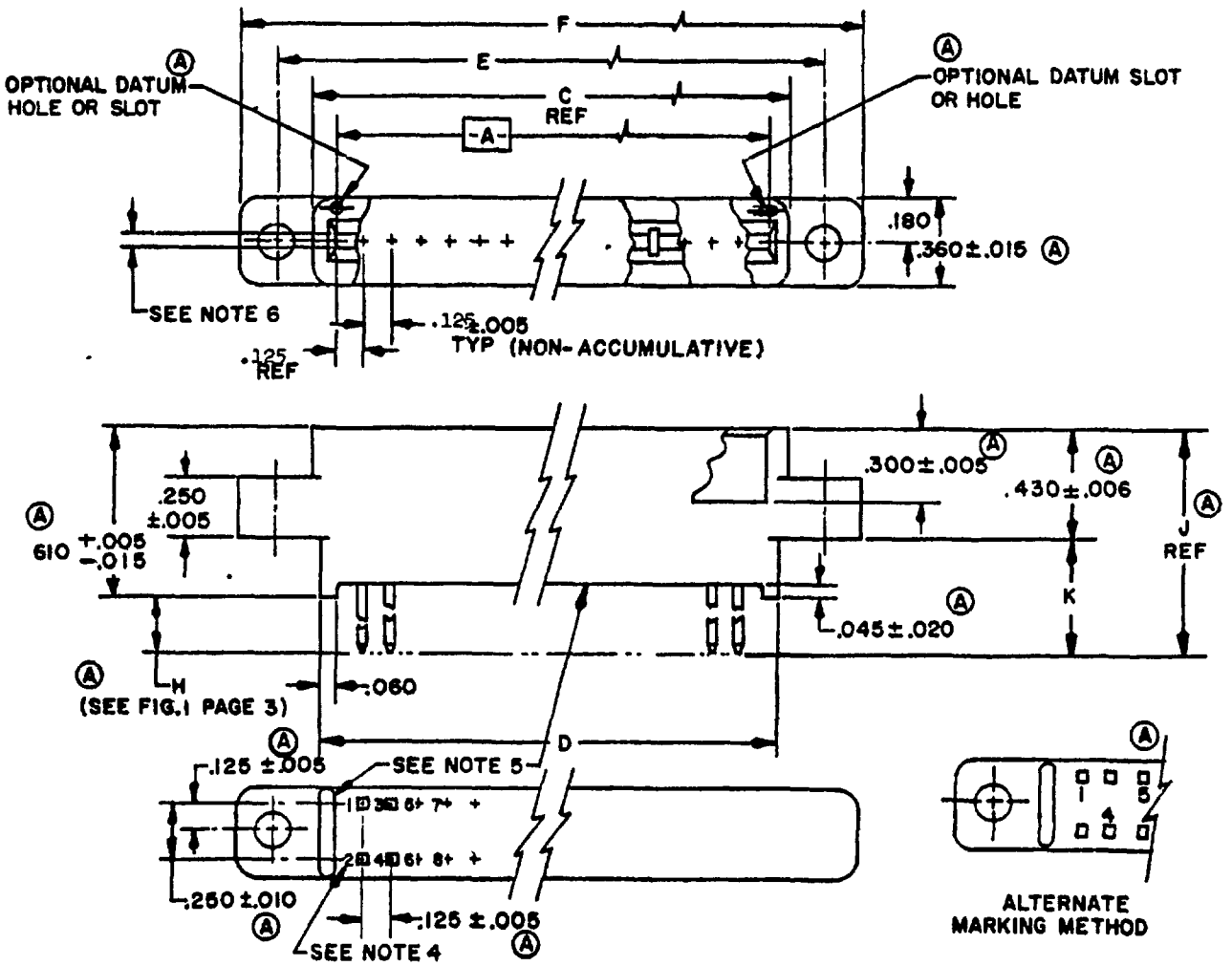
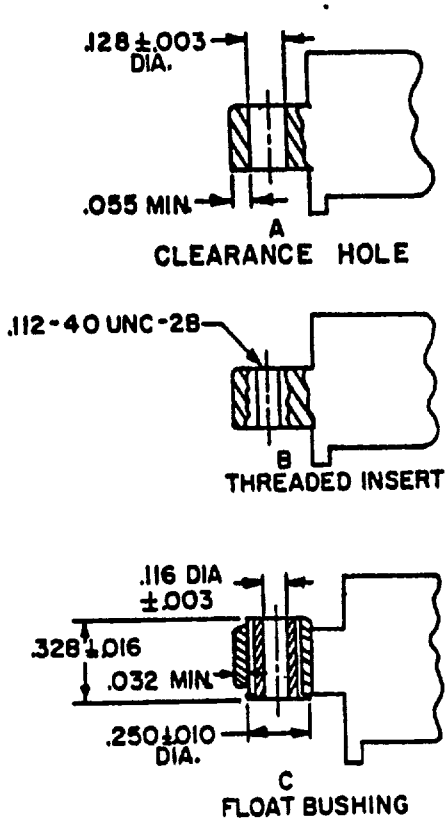


FIGURE 1. Type AD connectors.

Ⓐ denotes changes

FSC 5935

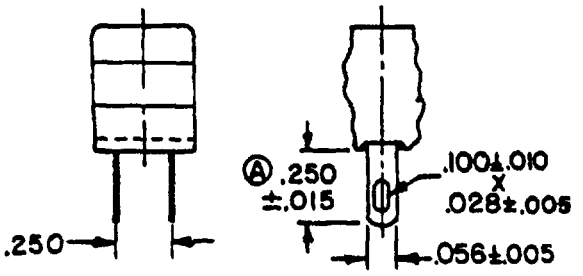
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.



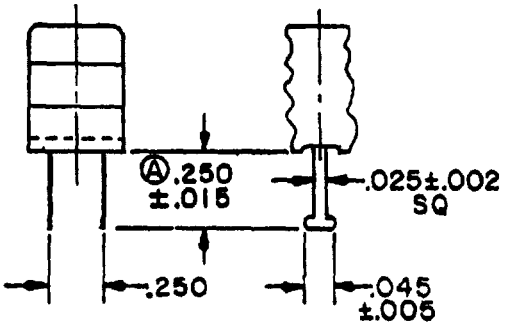
MOUNTING PROVISIONS

FIGURE 1. Type AD connectors - Continued.

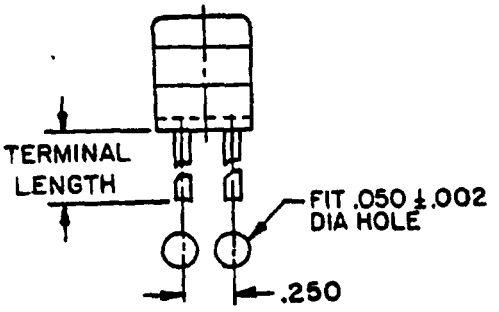
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Class 'a' dual contact, straight eyelet

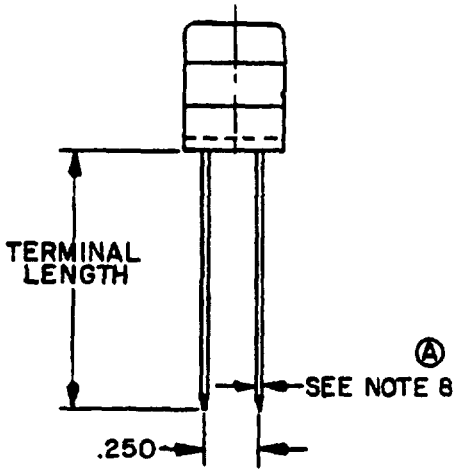


Optional



Class 'b' dual contact, straight plug-in

Mounting board thickness	Terminal length
.06	.109
.09	.146
.12	.177



Class 'd' wrappost

Ⓐ

No. of wire wraps	Terminal length	Wire size
2	.481 ±.020	26, 28, & 30 AWG
3	.665 ±.020	26, 28, & 30 AWG

TYPE AD DUAL CONTACT

FIGURE 1. Type AD connectors - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.025	.64	.109	2.77	.289	7.34	.665	16.89
.002	.05	.028	.71	.112	2.84	.300	7.62	.719	18.26
.003	.08	.032	.81	.116	2.95	.326	8.28	.756	19.20
.005	.13	.045	1.14	.12	3.05	.327	8.31	.787	19.99
.006	.15	.050	1.27	.125	3.18	.328	8.33	.845	21.46
.010	.25	.055	1.40	.128	3.25	.360	9.14	1.091	27.71
.015	.38	.056	1.42	.146	3.71	.430	10.92	1.275	32.39
.016	.41	.060	1.52	.177	4.50	.481	12.22	2.000	50.80
.020	.51	.09	2.29	.180	4.57	.610	15.49	2.160	54.86
				.250	6.35	.661	16.79	2.680	68.07

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.015 (.38 mm) for three place decimals and ±.02 (.5 mm) for two place decimals.
4. Contacts and terminals shall be designated on the connector using raised or silk screened numerals in the location shown on the connector drawing.
5. The shape of the barriers is optional except that barriers shall not protrude beyond the plane of the underside of the mounting flange. Center barrier shall be optional.
6. Slot shall accept the following thickness printed wiring board:

   Size 2 - .054 (1.37 mm) to .071 (1.80 mm).

   Slot shall prevent entry of a printed wiring board habing the following thicknesses:

   Size 2 - .084 (2.13 mm).
7. Gate allowance shall not exceed .005 (.13 mm) above or below the molded surface.
8. Ⓐ Class "d" wrappost geometry shall be in accordance with MIL-STD-1130. Wrappost contacts shall be gold plated in accordance with MIL-C-45204, type II grade C, class 1 over nickel plating per QQ-N-290, class 2, 30 microinches minimum.

FIGURE 1. Type AD connectors - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

Dash number	Type	Style	Class	Mounting provisions	±.007 A	C	D	±.010 E	±.020 F	H	J REF	±.030 K
001	AD	15	a	A	2.000	2.160	2.000	2.240	2.680	.250	.860	.430
002	↓	↓	a	B	↑	↑	↑	↑	↑	.250	.860	.430
003	↓	↓	a	C	↓	↓	↓	↓	↓	.250	.860	.430
004	↓	↓	b	A	↓	↓	↓	↓	↓	.109	.719	.289
005	↓	↓	b	B	↓	↓	↓	↓	↓	.109	.719	.289
006	↓	↓	b	C	↓	↓	↓	↓	↓	.109	.719	.289
007	↓	↓	b	A	↓	↓	↓	↓	↓	.146	.756	.326
008	↓	↓	b	B	↓	↓	↓	↓	↓	.146	.756	.326
009	↓	↓	b	C	↓	↓	↓	↓	↓	.146	.756	.326
010	↓	↓	b	A	↓	↓	↓	↓	↓	.177	.787	.357
011	↓	↓	b	B	↓	↓	↓	↓	↓	.177	.787	.357
012	↓	↓	b	C	↓	↓	↓	↓	↓	.177	.787	.357
013	↓	↓	d	A	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
014	↓	↓	d	B	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
015	↓	↓	d	C	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
016	↓	↓	d	A	↓	↓	↓	↓	↓	.665 ±.020	1.275	.845
017	↓	↓	d	B	↓	↓	↓	↓	↓	.665 ±.020	1.275	.845
018	↓	↓	d	C	2.000	2.160	2.000	2.420	2.680	.665 ±.020	1.275	.845

Figure 1. Type AD connectors - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

Dash number	Type	Style	Class	Mounting provisions	±.007 A	C	D	±.010 E	±.020 F	H	J REF	±.030 K
019	AD	18	a	A	2.375	2.535	2.375	2.795	3.055	.250	.860	.430
020	↓	↓	a	B	↑	↑	↑	↑	↑	.250	.860	.430
021	↓	↓	a	C	↑	↑	↑	↑	↑	.250	.860	.430
022	↓	↓	b	A	↑	↑	↑	↑	↑	.109	.719	.289
023	↓	↓	b	B	↑	↑	↑	↑	↑	.109	.719	.289
024	↓	↓	b	C	↑	↑	↑	↑	↑	.109	.719	.289
025	↓	↓	b	A	↑	↑	↑	↑	↑	.146	.756	.326
026	↓	↓	b	B	↑	↑	↑	↑	↑	.146	.756	.326
027	↓	↓	b	C	↑	↑	↑	↑	↑	.146	.756	.326
028	↓	↓	b	A	↑	↑	↑	↑	↑	.177	.787	.357
029	↓	↓	b	B	↑	↑	↑	↑	↑	.177	.787	.357
030	↓	↓	b	C	↑	↑	↑	↑	↑	.177	.787	.357
031	↓	↓	d	A	↑	↑	↑	↑	↑	.481 ±.020	1.091	.661
032	↓	↓	d	B	↑	↑	↑	↑	↑	.481 ±.020	1.091	.661
033	↓	↓	d	C	↑	↑	↑	↑	↑	.481 ±.020	1.091	.661
034	↓	↓	d	A	↑	↑	↑	↑	↑	.665 ±.020	1.275	.845
035	↓	↓	d	B	↑	↓	↓	↓	↓	.665 ±.020	1.275	.845
036	↓	↓	d	C	2.375	2.535	2.375	2.795	3.055	.665 ±.020	1.275	.845

FIGURE 1 Type AD connectors - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

Dash number	Type	Style	Class	Mounting provisions	±.007 A	C	D	±.010 E	±.020 F	H	J REF	±.030 K
037	AD	22	a	A	2.875	3.035	2.875	3.295	3.555	.250	.860	.430
038			a	B						.250	.860	.430
039			a	C						.250	.860	.430
040			b	A						.109	.719	.289
041			b	B						.109	.719	.289
042			b	C						.109	.719	.289
043			b	A						.146	.756	.326
044			b	B						.146	.756	.326
045			b	C						.146	.756	.326
046			b	A						.177	.787	.357
047			b	B						.177	.787	.357
048			b	C						.177	.787	.357
049			d	A						.481 ±.020	1.091	.661
050			d	B						.481 ±.020	1.091	.661
051			d	C						.481 ±.020	1.091	.661
052			d	A						.665 ±.020	1.275	.845
053			d	B						.665 ±.020	1.275	.845
054			d	C	2.875	3.035	2.875	3.295	3.555	.665 ±.020	1.275	.845

FIGURE 1. Type AD connectors - Continued.

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Check the source to verify that this is the current version before use.

Dash number	Type	Style	Class	Mounting provisions	±.007 A	C	D	±.010 E	±.020 F	H	J REF	±.030 K
055	AD	25	a	A	3.250	3.410	3.250	3.670	3.930	.250	.860	.430
056	↑	↑	a	B	↑	↑	↑	↑	↑	.250	.860	.430
057	↑	↑	a	C	↑	↑	↑	↑	↑	.250	.860	.430
058	↑	↑	b	A	↑	↑	↑	↑	↑	.109	.719	.289
059	↑	↑	b	B	↑	↑	↑	↑	↑	.109	.719	.289
060	↑	↑	b	C	↑	↑	↑	↑	↑	.109	.719	.289
061	↑	↑	b	A	↑	↑	↑	↑	↑	.146	.756	.326
062	↑	↑	b	B	↑	↑	↑	↑	↑	.146	.756	.326
063	↑	↑	b	C	↑	↑	↑	↑	↑	.146	.756	.326
064	↑	↑	b	A	↑	↑	↑	↑	↑	.177	.787	.357
065	↑	↑	b	B	↑	↑	↑	↑	↑	.177	.787	.357
066	↑	↑	b	C	↑	↑	↑	↑	↑	.177	.787	.357
067	↑	↑	d	A	↑	↑	↑	↑	↑	.481 ±.020	1.091	.661
068	↑	↑	d	B	↑	↑	↑	↑	↑	.481 ±.020	1.091	.661
069	↑	↑	d	C	↑	↑	↑	↑	↑	.481 ±.020	1.091	.661
070	↑	↑	d	A	↑	↑	↑	↑	↑	.665 ±.020	1.275	.845
071	↑	↑	d	B	↑	↑	↑	↑	↑	.665 ±.020	1.275	.845
072	↓	↓	d	C	3.250	3.410	3.250	3.670	3.930	.665 ±.020	1.275	.845

FIGURE 1. Type AD connectors - Continued

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Dash number	Type	Style	Class	Mounting provisions	±.007 A	C	D	±.010 E	±.020 F	H	J REF	±.030 K
073	AD	30	a	A	3.875	4.035	3.875	4.295	4.555	.250	.860	.430
074	↓	↓	a	B	↑	↑	↑	↑	↑	.250	.860	.430
075	↓	↓	a	C	↓	↓	↓	↓	↓	.250	.860	.430
076	↓	↓	b	A	↓	↓	↓	↓	↓	.109	.719	.289
077	↓	↓	b	B	↓	↓	↓	↓	↓	.109	.719	.289
078	↓	↓	b	C	↓	↓	↓	↓	↓	.109	.719	.289
079	↓	↓	b	A	↓	↓	↓	↓	↓	.146	.756	.326
080	↓	↓	b	B	↓	↓	↓	↓	↓	.146	.756	.326
081	↓	↓	b	C	↓	↓	↓	↓	↓	.146	.756	.326
082	↓	↓	b	A	↓	↓	↓	↓	↓	.177	.787	.357
083	↓	↓	b	B	↓	↓	↓	↓	↓	.177	.787	.357
084	↓	↓	b	C	↓	↓	↓	↓	↓	.177	.787	.357
085	↓	↓	d	A	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
086	↓	↓	d	B	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
087	↓	↓	d	C	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
088	↓	↓	d	A	↓	↓	↓	↓	↓	.665 ±.020	1.275	.845
089	↓	↓	d	B	↓	↓	↓	↓	↓	.665 ±.020	1.275	.845
090	↓	↓	d	C	3.875	4.035	3.875	4.295	4.555	.665 ±.020	1.275	.845

FIGURE 1. Type AD connectors - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
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Dash number	Type	Style	Class	Mounting provisions	±.007 A	C	D	±.010 E	±.020 F	H	J REF	±.030 K
091	AD	36	a	A	4.625	4.785	4.625	5.045	5.305	.250	.860	.430
092	↓	↓	a	B	↑	↑	↑	↑	↑	.250	.860	.430
093	↓	↓	a	C	↓	↓	↓	↓	↓	.250	.860	.430
094	↓	↓	b	A	↓	↓	↓	↓	↓	.109	.719	.289
095	↓	↓	b	B	↓	↓	↓	↓	↓	.109	.719	.289
096	↓	↓	b	C	↓	↓	↓	↓	↓	.109	.719	.289
097	↓	↓	b	A	↓	↓	↓	↓	↓	.146	.756	.326
098	↓	↓	b	B	↓	↓	↓	↓	↓	.146	.756	.326
099	↓	↓	b	C	↓	↓	↓	↓	↓	.146	.756	.326
100	↓	↓	b	A	↓	↓	↓	↓	↓	.177	.787	.357
101	↓	↓	b	B	↓	↓	↓	↓	↓	.177	.787	.357
102	↓	↓	b	C	↓	↓	↓	↓	↓	.177	.787	.357
103	↓	↓	d	A	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
104	↓	↓	d	B	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
105	↓	↓	d	C	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
106	↓	↓	d	A	↓	↓	↓	↓	↓	.665 ±.020	1.275	.845
107	↓	↓	d	B	↓	↓	↓	↓	↓	.665 ±.020	1.275	.845
108	↓	↓	d	C	4.625	4.785	4.625	5.045	5.305	.665 ±.020	1.275	.845

FIGURE 1. Type AD connectors - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
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Dash number	Type	Style	Class	Mounting provisions	±.010 A	C	D	±.015 E	±.020 F	H	J REF	±.030 K
109	AD	40	a	A	5.125	5.285	5.125	5.545	5.805	.250	.860	.430
110	↓	↓	a	B	↑	↑	↑	↑	↑	.250	.860	.430
111	↓	↓	a	C	↑	↑	↑	↑	↑	.250	.860	.430
112	↓	↓	b	A	↑	↑	↑	↑	↑	.109	.719	.289
113	↓	↓	b	B	↑	↑	↑	↑	↑	.109	.719	.289
114	↓	↓	b	C	↑	↑	↑	↑	↑	.109	.719	.289
115	↓	↓	b	A	↑	↑	↑	↑	↑	.146	.756	.326
116	↓	↓	b	B	↑	↑	↑	↑	↑	.146	.756	.326
117	↓	↓	b	C	↑	↑	↑	↑	↑	.146	.756	.326
118	↓	↓	b	A	↑	↑	↑	↑	↑	.177	.787	.357
119	↓	↓	b	B	↑	↑	↑	↑	↑	.177	.787	.357
120	↓	↓	b	C	↑	↑	↑	↑	↑	.177	.787	.357
121	↓	↓	d	A	↑	↑	↑	↑	↑	.481 ±.020	1.091	.661
122	↓	↓	d	B	↑	↑	↑	↑	↑	.481 ±.020	1.091	.661
123	↓	↓	d	C	↑	↑	↑	↑	↑	.481 ±.020	1.091	.661
124	↓	↓	d	A	↑	↑	↑	↑	↑	.665 ±.020	1.275	.845
125	↓	↓	d	B	↓	↓	↓	↓	↓	.665 ±.020	1.275	.845
126	↓	↓	d	C	5.125	5.285	5.125	5.545	5.805	.665 ±.020	1.275	.845

FIGURE 1. Type AD connectors - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

Dash number	Type	Style	Class	Mounting provisions	±.010 A			±.015 E	±.020 F	H	J REF	±.030 K
127	AD	43	a	A	5.500	5.660	5.500	5.920	6.180	.250	.860	.430
128	↓	↓	a	B	↑	↑	↑	↑	↑	.250	.860	.430
129			a	C						.250	.860	.430
130			b	A						.109	.719	.289
131			b	B						.109	.719	.289
132			b	C						.109	.719	.289
133			b	A						.146	.756	.326
134			b	B						.146	.756	.326
135			b	C						.146	.756	.326
136			b	A						.177	.787	.357
137			b	B						.177	.787	.357
138			b	C						.177	.787	.357
139			d	A						.481 ±.020	1.091	.661
140			d	B						.481 ±.020	1.091	.661
141			d	C						.481 ±.020	1.091	.661
142			d	A						.665 ±.020	1.275	.845
143			d	B	↓	↓	↓	↓	↓	.665 ±.020	1.275	.845
144	↓	↓	d	C	5.500	5.660	5.500	5.920	6.180	.665 ±.020	1.275	.845

FIGURE 1. Type AD connectors - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

Dash number	Type	Style	Class	Mounting provisions	±.010 A	C	D	±.015 E	±.020 F	H	J REF	±.030 K
145	AD	50	a	A	6.375	6.535	6.385	6.795	7.055	.250	.860	.430
146	↓	↓	a	B	↑	↑	↑	↑	↑	.250	.860	.430
147	↓	↓	a	C	↓	↓	↓	↓	↓	.250	.860	.430
148	↓	↓	b	A	↓	↓	↓	↓	↓	.109	.719	.289
149	↓	↓	b	B	↓	↓	↓	↓	↓	.109	.719	.289
150	↓	↓	b	C	↓	↓	↓	↓	↓	.109	.719	.289
151	↓	↓	b	A	↓	↓	↓	↓	↓	.146	.756	.326
152	↓	↓	b	B	↓	↓	↓	↓	↓	.146	.756	.326
153	↓	↓	b	C	↓	↓	↓	↓	↓	.146	.756	.326
154	↓	↓	b	A	↓	↓	↓	↓	↓	.177	.787	.357
155	↓	↓	b	B	↓	↓	↓	↓	↓	.177	.787	.357
156	↓	↓	b	C	↓	↓	↓	↓	↓	.177	.787	.357
157	↓	↓	d	A	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
158	↓	↓	d	B	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
159	↓	↓	d	C	↓	↓	↓	↓	↓	.481 ±.020	1.091	.661
160	↓	↓	d	A	↓	↓	↓	↓	↓	.665 ±.020	1.275	.845
161	↓	↓	d	B	↓	↓	↓	↓	↓	.665 ±.020	1.275	.845
162	↓	↓	d	C	6.375	6.535	6.385	6.795	7.055	.665 ±.020	1.275	.845

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Polarizing key slots: For contact polarization, slots shall be centered between contact positions, and shall accept keys supplied with the connector. Five keys shall be supplied with each connector. When fully seated into connector key slot the dimension which key extends above the key slot is .025 inch max. The thickness of the key shall be .026 ±.008 inch.

Current rating: 3 amperes.

Total engagement force (styles 40 and 50 only):

No. of contact positions	Max. board insertion force (gage)
40	51
50	63

Part number: M21097/24- (dash number from figure 1).

Qualification and group B inspection: When connector assemblies covering a range of styles, types, mountings and classes with the same design are submitted for inspection at the same time, the requirements of table I apply.

TABLE I. Group submission.

Dash number 1/	Samples (Inspection per tables IX and XII in MIL-C-21097)			
	Group I	Group II	Group III	Group IV
145	1		1	
149	1		1	
157		1		1 3/
162		1		1 3/
003	2			
013		2 2/		
022	2			
035		2 2/		
045	2			
051		2 2/		
065	2			
072		2 2/		
074	2			
084		2 2/		
091	2			
104		2 2/		
116	2			
125		2 2/		
129	2			
144		2 2/		

1/ Table shows sample selection for size 2, style 15 (shortest) connector to style 50 (longest) connector. For test programs not including the full range of styles or covering features of type or size not covered by this table, the test program shall be established in accordance with the guidelines reflected in this table.

2/ Only the following tests of group II are applicable:
Visual and mechanical examination
Board insertion force
Durability

3/ Not applicable for group B inspection.

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Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - MI
Navy - AS
Air Force - 11
DLA - ES

User activities:
Army - AR ME, AV, AT,
Navy - SH, MC, OS
Air Force -

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5935-3023-3)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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